



# General sum-connectivity index of unicyclic graphs with given maximum degree

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## ABSTRACT

For  $a \in \mathbb{R}$ , the general sum-connectivity index  $\chi_a$  of a graph  $G$  is defined as  $\chi_a(G) = \sum_{uv \in E(G)} [d_G(u) + d_G(v)]^a$ , where  $E(G)$  is the set of edges of  $G$ , and  $d_G(u)$  and  $d_G(v)$  are the degrees of vertices  $u$  and  $v$ , respectively. Among unicyclic graphs with given number of vertices and maximum degree, we present graphs having the largest and smallest values of  $\chi_a$ , and we state cases which are still open. We also solve one of the open problems on  $\chi_a$  for trees if  $0 < a < 1$ .

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## 1. Introduction

Graph theory is often used to investigate chemical structures, since atoms of molecules can be represented by vertices of graphs and chemical bonds can be represented by edges of graphs. We denote the set of vertices of a graph  $G$  by  $V(G)$  and the set of edges by  $E(G)$ . The number of vertices of  $G$  is the order  $n$  of  $G$ . The number of vertices adjacent to a vertex  $v$  in  $G$  is the degree of  $v$ , denoted by  $d_G(v)$ . The degree of a vertex whose degree is the largest in  $G$  is the maximum degree  $\Delta$  of  $G$ . A path such that the degree of each internal vertex (if any) is 2, the degree of one end vertex is greater than 2 and the degree of the other end vertex is 1 in  $G$ , is a pendant path of  $G$ . A vertex having degree one is a pendant vertex. A connected graph not containing cycles is a tree. A connected graph containing exactly one cycle is a unicyclic graph. We denote by  $C_k = v_1 v_2 \dots v_k v_1$  the cycle with  $k$  vertices  $v_1, v_2, \dots, v_k$  and edges  $v_1 v_2, v_2 v_3, \dots, v_{k-1} v_k, v_k v_1$ .

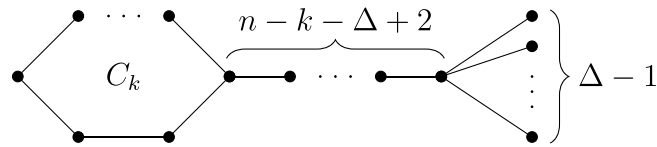
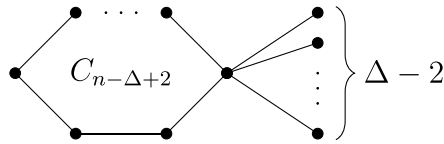
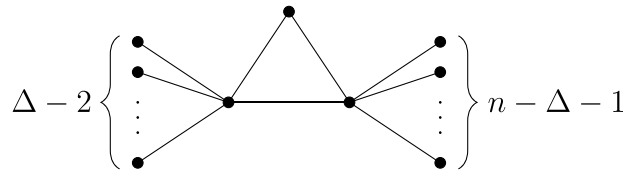
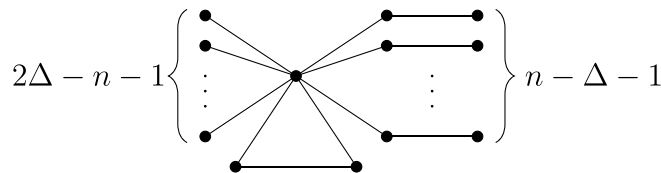
Indices of graphs have been studied due to their extensive applications, particularly in chemistry. The general sum-connectivity index is one of the most well-known general indices. For a graph  $G$  and  $a \in \mathbb{R}$ , it was defined by Zhou and Trinajstić [13] as

$$\chi_a(G) = \sum_{uv \in E(G)} [d_G(u) + d_G(v)]^a.$$

Unicyclic graphs are important networks which can represent chemical structures. We present extremal results on  $\chi_a$  for unicyclic graphs. For unicyclic graphs with prescribed order, graphs having extremal  $\chi_a$  were given by Du, Zhou and Trinajstić [4]. For  $-1 \leq a < 0$ , lower bounds on  $\chi_a$  for unicyclic graphs with given order and one other invariant were given in [6,10,11]. That other invariant is the matching number in [6], the number of pendant vertices in [10] and the girth in [11]. In [6], only the values of  $a$  such that  $-0.585 \leq a < 0$  are considered. Among unicyclic graphs with prescribed order and independence number, Tomescu [9] obtained the graph having the largest  $\chi_a$  for  $a \geq 1$ . Bounds on  $\chi_a$  for unicyclic graphs with prescribed order and diameter were given in [1]. For graphs with given cyclomatic number, Ali et al. [2] obtained extremal graphs on  $\chi_a$  if  $a > 1$ . A survey paper on  $\chi_a$  was produced by Ali, Zhong and Gutman [3].

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Fig. 1. Graph  $CPS_{k, n-k-\Delta+2, \Delta-1}$ .Fig. 2. Graph  $CS_{n-\Delta+2, \Delta-2}$ .Fig. 3. Graph  $CSS_{3, \Delta-2, n-\Delta-1}$ .Fig. 4. Graph  $CSI_{3, 2\Delta-n-1, n-\Delta-1}$ .

Let us define unicyclic graphs having extremal  $\chi_a$  in terms of the order  $n$  and maximum degree  $\Delta$ . In the notations below, “C” indicates that a graph contains a cycle as a subgraph, “P” indicates that a graph contains a path, “S” indicates that a graph contains a star and “I” indicates that a graph contains pendant paths of length at least 2.

For  $\Delta \geq 3$  and  $3 \leq k \leq n - \Delta - 1$ , let  $CPS_{k, n-k-\Delta+2, \Delta-1}$  be the graph obtained from the path of length  $n - k - \Delta + 1$  by identifying one of its end vertices with one vertex of the cycle  $C_k$  and by joining the other end vertex of that path to  $\Delta - 1$  new vertices; see Fig. 1.

For  $3 \leq \Delta \leq n - 4$ , we define the set  $CPS_{\Delta-1}^n$  as

$$CPS_{\Delta-1}^n = \{CPS_{k, n-k-\Delta+2, \Delta-1} : 3 \leq k \leq n - \Delta - 1\} = \{CPS_{3, n-\Delta-1, \Delta-1}, CPS_{4, n-\Delta-2, \Delta-1}, \dots, CPS_{n-\Delta-1, 3, \Delta-1}\}.$$

Every graph in  $CPS_{\Delta-1}^n$  has  $n$  vertices and maximum degree  $\Delta$ . The set  $CPS_{\Delta-1}^n$  contains  $n - \Delta - 3$  graphs. Note that if  $\Delta = n - 4$ , then  $CPS_{\Delta-1}^n = \{CPS_{3, 3, n-5}\}$  which means that  $CPS_{\Delta-1}^n$  contains only one graph.

For  $3 \leq \Delta \leq n - 1$ , let  $CS_{n-\Delta+2, \Delta-2}$  be the graph obtained from the cycle  $C_{n-\Delta+2}$  by joining one of its vertices to  $\Delta - 2$  new vertices; see Fig. 2.

For  $\lceil \frac{n+1}{2} \rceil \leq \Delta \leq n - 1$ , let  $CSS_{3, \Delta-2, n-\Delta-1}$  be the graph obtained from the cycle  $C_3$  by joining one of its vertices to  $\Delta - 2$  new vertices, and by joining one other vertex of that  $C_3$  to  $n - \Delta - 1$  new vertices; see Fig. 3.

For  $\lceil \frac{n+1}{2} \rceil \leq \Delta \leq n - 1$ , we denote a graph obtained by attaching  $2\Delta - n - 1$  pendant paths having length 1 and  $n - \Delta - 1$  pendant paths having length 2 to the same vertex of the cycle  $C_3$  by  $CSI_{3, 2\Delta-n-1, n-\Delta-1}$ ; see Fig. 4.

For  $3 \leq \Delta \leq \lfloor \frac{n+1}{2} \rfloor$ , the set  $CI_{\Delta-2}^n$  contains each graph of order  $n$  obtained by attaching  $\Delta - 2$  pendant paths of length at least 2 to the same vertex of a cycle. For  $3 \leq \Delta \leq n - 1$ , the set  $CV_{\Delta-2}^n$  contains every graph with  $n$  vertices obtained by attaching  $\Delta - 2$  pendant paths (of length at least 1) to the same vertex of a cycle.

For any unicyclic graph with  $n$  vertices and maximum degree  $\Delta$ , we have  $2 \leq \Delta \leq n - 1$ . The only unicyclic graph with  $n$  vertices and maximum degree 2, where  $n \geq 3$ , is  $C_n$ , and the only unicyclic graph with  $n$  vertices and maximum degree  $n - 1$ , where  $n \geq 4$ , is  $CS_{3, n-3}$ . Therefore we present known and our results on unicyclic graphs with given  $n$  and  $\Delta$ , where  $3 \leq \Delta \leq n - 2$ .

For  $-1 \leq a \leq 0$ , Jamil and Tomescu [7] proved that among unicyclic graphs with given  $n$  and  $\Delta$ , the graph  $CS_{3,2\Delta-n-1,n-\Delta-1}$  has the largest  $\chi_a$  for  $\lceil \frac{n+1}{2} \rceil \leq \Delta \leq n-2$  and each graph in the set  $CI_{\Delta-2}^n$  has the largest  $\chi_a$  for  $3 \leq \Delta \leq \lfloor \frac{n+1}{2} \rfloor$ . Horoldagva and Das [5] showed that every graph in the set  $CY_{\Delta-2}^n$  has the smallest  $\chi_a$  if  $a = 1$ .

Our results on unicyclic graphs are given in Theorems 2.1, 3.1 and 4.1. For unicyclic graphs with prescribed  $n$  and  $\Delta$ , graphs having the smallest  $\chi_a$  for  $a > 0$  are given in Theorem 2.1, the graph having the largest  $\chi_a$  for  $a < 0$  is given in Theorem 3.1, and the graph with the smallest  $\chi_a$  for  $a < 0$  and largest  $\chi_a$  for  $a > 0$  is given in Theorem 4.1. Theorems 3.1 and 4.1 are presented for  $\lceil \frac{n+1}{2} \rceil \leq \Delta \leq n-2$ . In Problems 5.1 and 5.2, we state cases which are still open for unicyclic graphs. The tree with given  $n$  and  $\Delta$  having the largest  $\chi_a$  for  $0 < a < 1$  and  $\lceil \frac{n}{2} \rceil \leq \Delta \leq n-2$  is given in Theorem 4.2.

We include the first lemma in this section, because it is used in all three sections (Sections 2–4) containing our theorems. Lemma 1.1 was proved in [12]. The proof is short, thus we include it.

**Lemma 1.1.** *Let  $1 \leq x_1 < x_2$  and  $c > 0$ . Then for  $a > 1$  and  $a < 0$ ,*

$$(x_1 + c)^a - x_1^a < (x_2 + c)^a - x_2^a.$$

*For  $0 < a < 1$ ,*

$$(x_1 + c)^a - x_1^a > (x_2 + c)^a - x_2^a.$$

**Proof.** Let  $f(x) = (x + c)^a - x^a$ , where  $x \geq 1$ . The derivative

$$f'(x) = a[(x + c)^{a-1} - x^{a-1}].$$

If  $a > 1$ , then  $(x + c)^{a-1} > x^{a-1}$ . If  $a < 1$ , then  $(x + c)^{a-1} < x^{a-1}$ .

Thus, for  $a > 1$  and  $a < 0$ , we have  $f'(x) > 0$ , so  $f(x)$  is strictly increasing. Hence, for  $1 \leq x_1 < x_2$ , we obtain  $f(x_1) < f(x_2)$ .

For  $0 < a < 1$ ,  $f'(x) < 0$ , so  $f(x)$  is strictly decreasing. Hence, for  $1 \leq x_1 < x_2$ , we obtain  $f(x_1) > f(x_2)$ .  $\square$

## 2. Smallest value of $\chi_a$ for $a > 0$

We present several lemmas. Lemma 2.1 was proved in [8]. Since the proof is very short, we include it.

**Lemma 2.1.** *Let  $a, c, q, r \in \mathbb{R}$ , such that  $c, q, r > 0$  and  $\{q, r\} \neq \{1\}$ . Then the function*

$$f_{c,q,r}(a) = c q^a + r^a$$

*is strictly convex.*

**Proof.** The second derivative

$$f''_{c,q,r}(a) = (\ln q)^2 c q^a + (\ln r)^2 r^a > 0,$$

so  $f_{c,q,r}(a)$  is strictly convex.  $\square$

Let us present Lemma 2.2 which is used in the proofs of Lemma 2.3 and Theorem 2.1.

**Lemma 2.2.** *We have*

- (i)  $2(5^a) + 3^a - 3(4^a) > 0$  for  $a > 0$ ,
- (ii)  $6^a + 3^a - 2(4^a) > 0$  for  $a > 0$ .

**Proof.** For  $a \in \mathbb{R}$ , the functions

$$f_{2, \frac{5}{4}, \frac{3}{4}}(a) = 2 \left( \frac{5}{4} \right)^a + \left( \frac{3}{4} \right)^a \quad \text{and} \quad f_{1, \frac{3}{2}, \frac{3}{4}}(a) = \left( \frac{3}{2} \right)^a + \left( \frac{3}{4} \right)^a$$

are strictly convex by Lemma 2.1.

- (i) We have  $f_{2, \frac{5}{4}, \frac{3}{4}}(-1) = \frac{44}{15}$  and  $f_{2, \frac{5}{4}, \frac{3}{4}}(0) = 3$ , therefore  $f_{2, \frac{5}{4}, \frac{3}{4}}(a) > 3$  for  $a > 0$ . So,  $2 \left( \frac{5}{4} \right)^a + \left( \frac{3}{4} \right)^a > 3$ , thus  $2(5^a) + 3^a > 3(4^a)$ .
- (ii) We have  $f_{1, \frac{3}{2}, \frac{3}{4}}(-1) = f_{1, \frac{3}{2}, \frac{3}{4}}(0) = 2$ . Since  $f_{1, \frac{3}{2}, \frac{3}{4}}(a)$  is strictly convex, we get  $f_{1, \frac{3}{2}, \frac{3}{4}}(a) > 2$  for  $a > 0$ . So  $\left( \frac{3}{2} \right)^a + \left( \frac{3}{4} \right)^a > 2$ , thus  $6^a + 3^a > 2(4^a)$  for  $a > 0$ .  $\square$

In Lemma 2.3, we compare  $\chi_a$  of two graphs for  $a > 0$ . Note that if the graph  $G_1$  is unicyclic, then  $G_2$  is also unicyclic.

**Lemma 2.3.** Let  $G_1$  be a graph containing a pendant path with end vertices  $u_0$  and  $u_p$ , where  $3 \leq d_{G_1}(u_0) \leq 4$  and  $d_{G_1}(u_p) = 1$ . Let  $u'$  be one of the neighbours of  $u_0$  in  $G_1$  which is not on that path, where

$$(d_{G_1}(u_0), d_{G_1}(u')) = (3, 3), (3, 2) \text{ or } (4, 2).$$

We define  $G_2$  with  $V(G_2) = V(G_1)$  and  $E(G_2) = \{u_p u'\} \cup E(G_1) \setminus \{u_0 u'\}$ . Then for  $a > 0$ , we have  $\chi_a(G_2) < \chi_a(G_1)$ .

**Proof.** We prove the result for  $d_{G_1}(u_0) = 3$ . (The proof for  $d_{G_1}(u_0) = 4$  is similar.) Let  $u'$  and  $u''$  be the neighbours of  $u_0$  in  $G_1$  which are not on the path between  $u_0$  and  $u_p$ . Let  $p$  be the length of the pendant path containing the end vertices  $u_0$  and  $u_p$ . If  $p = 1$ , then

$$\begin{aligned} \chi_a(G_1) - \chi_a(G_2) &= [d_{G_1}(u_0) + d_{G_1}(u')]^a - [d_{G_2}(u_1) + d_{G_2}(u')]^a + [d_{G_1}(u_0) + d_{G_1}(u'')]^a - [d_{G_2}(u_0) + d_{G_2}(u'')]^a \\ &\quad + [d_{G_1}(u_0) + d_{G_1}(u_1)]^a - [d_{G_2}(u_0) + d_{G_2}(u_1)]^a \\ &= [3 + d_{G_1}(u')]^a - [2 + d_{G_1}(u')]^a + [3 + d_{G_1}(u'')]^a - [2 + d_{G_1}(u'')]^a + (3 + 1)^a - (2 + 2)^a \\ &> 0. \end{aligned}$$

If  $p \geq 2$ , then

$$\begin{aligned} \chi_a(G_1) - \chi_a(G_2) &= [d_{G_1}(u_0) + d_{G_1}(u')]^a - [d_{G_2}(u_p) + d_{G_2}(u')]^a + [d_{G_1}(u_0) + d_{G_1}(u'')]^a - [d_{G_2}(u_0) + d_{G_2}(u'')]^a \\ &\quad + (d_{G_1}(u_0) + 2)^a - (d_{G_2}(u_0) + 2)^a + (d_{G_1}(u_p) + 2)^a - (d_{G_2}(u_p) + 2)^a \\ &= [3 + d_{G_1}(u')]^a - [2 + d_{G_1}(u')]^a + [3 + d_{G_1}(u'')]^a - [2 + d_{G_1}(u'')]^a \\ &\quad + (3 + 2)^a - (2 + 2)^a + (1 + 2)^a - (2 + 2)^a \\ &> [3 + d_{G_1}(u')]^a - [2 + d_{G_1}(u')]^a + 5^a - 2(4^a) + 3^a. \end{aligned}$$

If  $d_{G_1}(u') = 2$ , then  $\chi_a(G_1) - \chi_a(G_2) > 2(5^a) - 3(4^a) + 3^a > 0$  by Lemma 2.2(i). If  $d_{G_1}(u') = 3$ , then  $\chi_a(G_1) - \chi_a(G_2) > 6^a - 2(4^a) + 3^a > 0$  by Lemma 2.2(ii). So  $\chi_a(G_1) > \chi_a(G_2)$ .  $\square$

In Lemma 2.4, we compare  $\chi_a$  of two graphs such that one graph is obtained from the other graph by replacing two pendant paths with one pendant path. The proof of Lemma 2.4 was given in [13] in the proof of their Proposition 3.

**Lemma 2.4.** Let  $G_1$  be a graph containing a vertex  $v$  of degree at least 3 in  $G_1$ , such that  $v$  is an end vertex of (at least) two pendant paths having lengths  $r$  and  $s$ , respectively, where  $r, s \geq 1$ . Let  $G_2$  be obtained from  $G_1$  by replacing those two pendant paths by one pendant path of length  $r + s$ . Then  $\chi_a(G_1) > \chi_a(G_2)$  for  $a > 0$ .

In Theorem 2.1, we present unicyclic graphs having the smallest  $\chi_a$  for  $a > 0$ . From Theorem 2.1, it follows that if  $\Delta = \frac{n+1}{2}$ , then  $CI_{\Delta-2}^n = \{CS_{3,2\Delta-n-1,n-\Delta-1}\}$ .

**Theorem 2.1.** Among unicyclic graphs with order  $n$  and maximum degree  $\Delta$ , where  $3 \leq \Delta \leq n - 2$ , let  $X$  be the set of all graphs having the smallest value of  $\chi_a$ . For  $a > 1$  and  $n - 3 \leq \Delta \leq n - 2$ ,

$$X = \{CS_{n-\Delta+2,\Delta-2}\}.$$

For  $a > 1$  and  $3 \leq \Delta \leq n - 4$ ,

$$X = \begin{cases} CPS_{\Delta-1}^n & \text{if } (\Delta + 2)^a - (\Delta + 1)^a + 3(4^a - 5^a) > 0, \\ CPS_{\Delta-1}^n \cup \{CS_{n-\Delta+2,\Delta-2}\} & \text{if } (\Delta + 2)^a - (\Delta + 1)^a + 3(4^a - 5^a) = 0, \\ \{CS_{n-\Delta+2,\Delta-2}\} & \text{if } (\Delta + 2)^a - (\Delta + 1)^a + 3(4^a - 5^a) < 0. \end{cases}$$

For  $0 < a < 1$ ,

$$X = \begin{cases} CI_{\Delta-2}^n & \text{if } 3 \leq \Delta \leq \left\lfloor \frac{n+1}{2} \right\rfloor, \\ \{CS_{3,2\Delta-n-1,n-\Delta-1}\} & \text{if } \left\lceil \frac{n+1}{2} \right\rceil \leq \Delta \leq n - 2. \end{cases}$$

**Proof.** Among unicyclic graphs of order  $n$  and maximum degree  $\Delta$ , where  $3 \leq \Delta \leq n - 2$ , let  $U'$  be any unicyclic graph having the smallest  $\chi_a$ . Let  $C_k = v_1 v_2 \dots v_k v_1$  be the unique cycle in  $U'$ . Let  $w$  be a vertex of degree  $\Delta$  in  $U'$ . Let  $P$  be a path between  $w$  and a vertex on  $C_k$  which is closest to  $w$ . Without loss of generality, assume that  $v_1$  is closest to  $w$ . Note that the length of  $P$  can be 0 if  $w \in C_k$ . Let  $S$  be the set containing all the vertices of  $C_k$  and  $P$ , so  $S = V(C_k) \cup V(P)$ .

**Claim 1.** Each vertex which is not in  $S$  has degree at most 2.

Assume to the contrary that there exists a vertex which is not in  $S$  with degree at least 3. By  $v$ , we denote a vertex farthest from  $S$  such that  $d_{U'}(v) \geq 3$ . Then  $v$  is an end vertex of (at least) two pendant paths, thus by Lemma 2.4,  $U'$  does not have the smallest  $\chi_a$  which is a contradiction.

**Claim 2.** Each vertex in  $S \setminus \{w\}$  is adjacent only to vertices in  $S$ .

Assume to the contrary that there is a vertex in  $S \setminus \{w\}$ , say  $v$ , adjacent to  $r$  vertices not in  $S$ , where  $r \geq 1$ . From Claim 1,  $v$  is an end vertex of  $r$  pendant paths. The vertex  $v$  cannot be an end vertex of (at least) two pendant paths, because then by Lemma 2.4,  $U'$  would not have the smallest  $\chi_a$ . So  $r = 1$ . We cannot have  $v = v_i$  for  $2 \leq i \leq k$ , because we would have  $d_{U'}(v) = 3$  and  $v$  would have a neighbour on  $C$  of degree 2 or 3 in  $U'$ , thus by Lemma 2.3, there would be a unicyclic graph with smaller  $\chi_a$ . Consequently, we have  $v \neq v_1$ , otherwise we would have  $d_{U'}(v) = 4$  and  $v$  would have a neighbour on  $C$  of degree 2 in  $U'$ , thus by Lemma 2.3,  $U'$  would not have the smallest  $\chi_a$ . Finally,  $v \notin V(P) \setminus \{v_1, w\}$ , otherwise we would have  $d_{U'}(v) = 3$  and  $v$  would have a neighbour on  $P$  of degree 2 or 3 in  $U'$ , thus by Lemma 2.3, there would be a unicyclic graph with smaller  $\chi_a$ , a contradiction.

**Case 1.**  $a > 1$ .

**Claim 3.** All pendant paths containing an end vertex  $w$  have length 1.

Assume to the contrary that  $w$  is an end vertex of a pendant path with length  $p \geq 2$  in  $U'$ . Let us replace that path by the path of length 1, and we replace  $C_k$  by the cycle of length  $k + p - 1$  to obtain  $U_1$  from  $U'$ . Then

$$\chi_a(U') - \chi_a(U_1) = (\Delta + 2)^a - (\Delta + 1)^a + 3^a - 4^a > 0,$$

because  $(\Delta + 2)^a - (\Delta + 1)^a > 4^a - 3^a$  for  $a > 1$ , by Lemma 1.1. So  $\chi_a(U') > \chi_a(U_1)$  which is a contradiction.

**Claim 4.** The length of  $P$  is not 1.

Assume to the contrary that the length of  $P$  is 1. This means that  $v_1w \in E(U')$  and  $w$  is adjacent to  $\Delta - 1$  pendant vertices in  $U'$  (by Claim 3). Let  $w_1$  be any pendant vertex adjacent to  $w$ . We remove the edge  $v_1v_2$  from  $U'$  and add the edge  $v_2w_1$  to obtain a new unicyclic graph  $U_2$ . Then

$$\chi_a(U') - \chi_a(U_2) = [(\Delta + 3)^a - (\Delta + 2)^a] - [(\Delta + 2)^a - (\Delta + 1)^a] + 2(5^a - 4^a) > 0,$$

since  $5^a > 4^a$  and by Lemma 1.1, we have  $(\Delta + 3)^a - (\Delta + 2)^a > (\Delta + 2)^a - (\Delta + 1)^a$  for  $a > 1$ . So  $\chi_a(U') > \chi_a(U_2)$  which is a contradiction. The proof of Claim 4 is complete.

From Claims 2, 3 and 4, we know that the length of  $P$  is 0 or at least 2 which means that  $U'$  is  $CS_{n-\Delta+2, \Delta-2}$  or  $CPS_{k, n-k-\Delta+2, \Delta-1}$  for  $3 \leq k \leq n - \Delta - 1$ . From  $3 \leq n - \Delta - 1$ , we obtain  $\Delta \leq n - 4$  which means that  $CPS_{k, n-k-\Delta+2, \Delta-1}$  is not defined for  $n - 3 \leq \Delta \leq n - 2$ . Therefore  $U'$  is  $CS_{n-\Delta+2, \Delta-2}$  for  $n - 3 \leq \Delta \leq n - 2$ .

Let  $X$  be the set of all graphs having the smallest  $\chi_a$ . It follows that for  $n - 3 \leq \Delta \leq n - 2$ , we have  $X = \{CS_{n-\Delta+2, \Delta-2}\}$ .

Let  $3 \leq \Delta \leq n - 4$ . We compare  $\chi_a(CS_{n-\Delta+2, \Delta-2})$  and  $\chi_a(CPS_{k, n-k-\Delta+2, \Delta-1})$ , where  $3 \leq k \leq n - \Delta - 1$ . We have

$$\chi_a(CPS_{k, n-k-\Delta+2, \Delta-1}) = (\Delta + 2)^a + (\Delta - 1)(\Delta + 1)^a + 3(5^a) + (n - \Delta - 3)4^a$$

and

$$\chi_a(CS_{n-\Delta+2, \Delta-2}) = 2(\Delta + 2)^a + (\Delta - 2)(\Delta + 1)^a + (n - \Delta)4^a.$$

So

$$\chi_a(CS_{n-\Delta+2, \Delta-2}) - \chi_a(CPS_{n-\Delta-t+1, t, \Delta-1}) = (\Delta + 2)^a - (\Delta + 1)^a + 3(4^a - 5^a).$$

Hence  $CPS_{k, n-k-\Delta+2, \Delta-1}$  for any  $3 \leq k \leq n - \Delta - 1$  has the smallest  $\chi_a$  if  $(\Delta + 2)^a - (\Delta + 1)^a + 3(4^a - 5^a) > 0$ , so  $X = CPS_{\Delta-1}^n$ .

The graph  $CS_{n-\Delta+2, \Delta-2}$  has the smallest  $\chi_a$  if  $(\Delta + 2)^a - (\Delta + 1)^a + 3(4^a - 5^a) < 0$ , so  $X = \{CS_{n-\Delta+2, \Delta-2}\}$ .

The graphs  $CS_{n-\Delta+2, \Delta-2}$  and  $CPS_{k, n-k-\Delta+2, \Delta-1}$  for any  $3 \leq k \leq n - \Delta - 1$  have the smallest  $\chi_a$  if  $(\Delta + 2)^a - (\Delta + 1)^a + 3(4^a - 5^a) = 0$ , so  $X = CPS_{\Delta-1}^n \cup \{CS_{n-\Delta+2, \Delta-2}\}$ . The proof of Case 1 is complete.

**Case 2.**  $0 < a < 1$ .

**Claim 5.** The length of  $P$  is 0.

Assume to the contrary that  $w \neq v_1$ . Let  $wu_1u_2 \dots u_p$ , where  $p \geq 1$ , be any pendant path with an end vertex  $w$ . We remove the edge  $v_1v_2$  from  $U'$  and add the edge  $v_2u_p$  to obtain a new unicyclic graph  $U_3$ . We have

$$d_{U'}(u_p) = 1, d_{U_3}(u_p) = 2, d_{U'}(v_1) = 3, d_{U_3}(v_1) = d_{U'}(v_2) = d_{U_3}(v_2) = d_{U'}(v_k) = d_{U_3}(v_k) = 2.$$

We consider two cases:  $p = 1$  and  $p \geq 2$ .

Let  $p = 1$ . Then

$$\begin{aligned} \chi_a(U') - \chi_a(U_3) &= \sum_{u \in N_{U'}(v_1) \setminus \{v_2\}} ([d_{U'}(v_1) + d_{U'}(u)]^a - [d_{U_3}(v_1) + d_{U_3}(u)]^a) \\ &\quad + [d_{U'}(v_1) + d_{U'}(v_2)]^a - [d_{U_3}(v_2) + d_{U_3}(u_p)]^a + [d_{U'}(w) + d_{U'}(u_p)]^a - [d_{U_3}(w) + d_{U_3}(u_p)]^a \\ &= \sum_{u \in N_{U'}(v_1) \setminus \{v_2\}} ([3 + d_{U'}(u)]^a - [2 + d_{U'}(u)]^a) + (3 + 2)^a - (2 + 2)^a + (\Delta + 1)^a - (\Delta + 2)^a \\ &> 5^a - 4^a + (\Delta + 1)^a - (\Delta + 2)^a \\ &\geq 0, \end{aligned}$$

since  $[3 + d_{U'}(u)]^a > [2 + d_{U'}(u)]^a$ , and by Lemma 1.1, we have

$$5^a - 4^a \geq (\Delta + 2)^a - (\Delta + 1)^a.$$

Let  $p \geq 2$ . Then  $d_{U'}(u_{p-1}) = d_{U_3}(u_{p-1}) = 2$ . Let us denote the neighbour of  $v_1$  in  $U'$  which is not on  $C_k$  by  $u$ . We have

$$\begin{aligned} \chi_a(U') - \chi_a(U_3) &= [d_{U'}(v_1) + d_{U'}(u)]^a - [d_{U_3}(v_1) + d_{U_3}(u)]^a + [d_{U'}(v_1) + d_{U'}(v_k)]^a - [d_{U_3}(v_1) + d_{U_3}(v_k)]^a \\ &\quad + [d_{U'}(v_1) + d_{U'}(v_2)]^a - [d_{U_3}(v_2) + d_{U_3}(u_p)]^a + [d_{U'}(u_{p-1}) + d_{U'}(u_p)]^a - [d_{U_3}(u_{p-1}) + d_{U_3}(u_p)]^a \\ &= [3 + d_{U'}(u)]^a - [2 + d_{U'}(u)]^a + (3 + 2)^a - (2 + 2)^a + (3 + 2)^a - (2 + 2)^a + (2 + 1)^a - (2 + 2)^a \\ &> 2(5^a) - 3(4^a) + 3^a \\ &> 0, \end{aligned}$$

by Lemma 2.2(i). So  $\chi_a(U') > \chi_a(U_3)$  which means that  $U'$  does not have the smallest  $\chi_a$ . We have a contradiction. Hence  $w = v_1$ .

**Claim 6.** The pendant paths of  $U'$  either all are of length at most 2 or all are of length at least 2.

By contradiction, suppose that  $U'$  has a pendant path with length  $p \geq 3$  and a pendant path with length 1. We obtain a unicyclic graph  $U_4$  from  $U'$  by replacing those two pendant paths with pendant paths having lengths  $p - 1$  and 2. Then

$$\chi_a(U') - \chi_a(U_4) = (\Delta + 1)^a - (\Delta + 2)^a + 4^a - 3^a > 0,$$

since  $4^a - 3^a > (\Delta + 2)^a - (\Delta + 1)^a$  for  $0 < a < 1$ , by Lemma 1.1. Thus  $\chi_a(U') > \chi_a(U_4)$ , a contradiction.

**Claim 7.** If  $U'$  has a pendant path with length 1, then  $k = 3$ .

By contradiction, suppose that  $k \geq 4$  and  $U'$  has a pendant path with length 1. We obtain a new unicyclic graph  $U_5$  from  $U'$  by replacing the cycle  $C_k$  with the cycle of length  $k - 1$  and by replacing the pendant path of length 1 with a pendant path of length 2. Then

$$\chi_a(U') - \chi_a(U_5) = (\Delta + 1)^a - (\Delta + 2)^a + 4^a - 3^a > 0,$$

since  $4^a - 3^a > (\Delta + 2)^a - (\Delta + 1)^a$  by Lemma 1.1. Thus  $\chi_a(U') > \chi_a(U_5)$ , a contradiction.

**Case 2.1.**  $\lceil \frac{n+1}{2} \rceil \leq \Delta \leq n - 2$ .

We have  $\frac{n+1}{2} \leq \Delta$ , so  $n \leq 2\Delta - 1$ . Then  $U'$  does not contain a pendant path with length at least 3 (otherwise, if  $U'$  would have a pendant path with length at least 3, then by Claim 6, all the other  $\Delta - 3$  pendant paths would have lengths at least 2, and then  $U'$  would have at least  $2(\Delta - 3) + 3 + k \geq 2(\Delta - 3) + 3 + 3 = 2\Delta$  vertices, so  $n \geq 2\Delta$ , a contradiction). Thus, the length of each of the  $\Delta - 2$  pendant paths is 1 or 2.

If  $U'$  has a pendant path with length 1, then  $k = 3$  by Claim 7. If all the  $\Delta - 2$  pendant paths of  $U'$  have length 2, then again  $k = 3$  (otherwise if  $k \geq 4$ , then  $U'$  would have  $2(\Delta - 2) + k \geq 2\Delta$  vertices).

Let  $r$  be the number of pendant paths of length 2. Then  $w$  is adjacent to  $\Delta - 2 - r$  pendant vertices which implies that  $n = 3 + 2r + (\Delta - 2 - r) = \Delta + r + 1$  and consequently,  $r = n - \Delta - 1$ . So  $U'$  is  $CS_{3,2\Delta-n-1,n-\Delta-1}$  and  $X = \{CS_{3,2\Delta-n-1,n-\Delta-1}\}$ .

**Case 2.2.**  $3 \leq \Delta \leq \lfloor \frac{n+1}{2} \rfloor$ .

Then  $\Delta \leq \frac{n+1}{2}$ , so  $n \geq 2\Delta - 1$ . Then  $U'$  does not have a pendant path with length 1 (otherwise by Claim 7, the cycle of  $U'$  would be a  $C_3$ , and by Claim 6, all the other  $\Delta - 3$  pendant paths would have lengths at most 2, and then  $U'$  would have at most  $2(\Delta - 3) + 3 + 1 = 2\Delta - 2$  vertices, so  $n \leq 2\Delta - 2$ , a contradiction). Thus the length of each of the  $\Delta - 2$  pendant paths is at least 2. Hence  $X = C_{\Delta-2}^n$ .  $\square$

Note that for  $a > 2$  and  $12 \leq \Delta \leq n - 4$ , we have  $(\Delta + 2)^a - (\Delta + 1)^a + 3(4^a - 5^a) > 0$ , therefore in all such cases, every graph in  $CPS_{\Delta-1}^n$  has the smallest  $\chi_a$ .

For  $a > 1$  and  $\Delta = 3$ , we have  $(\Delta + 2)^a - (\Delta + 1)^a + 3(4^a - 5^a) < 0$ , so  $CS_{n-1,1}$  has the smallest  $\chi_a$ .

### 3. Largest value of $\chi_a$ for $a < 0$

In Theorem 3.1, we present the unicyclic graph having the largest  $\chi_a$  for  $a < 0$ .

**Theorem 3.1.** Among unicyclic graphs of order  $n$  and maximum degree  $\Delta$ , where  $\lceil \frac{n+1}{2} \rceil \leq \Delta \leq n - 2$ , the graph  $CS_{3,2\Delta-n-1,n-\Delta-1}$  has the largest value of  $\chi_a$  for  $a < 0$ .

**Proof.** Note that  $\lceil \frac{n+1}{2} \rceil \leq \Delta \leq n - 2$  gives  $n \geq 5$ , and then  $\Delta \geq 3$ . We also obtain  $n \leq 2\Delta - 1$ .

Among unicyclic graphs with maximum degree  $\Delta$  and order  $n$ , let  $U'$  be any unicyclic graph with the largest  $\chi_a$ . Let  $w$  be a vertex of degree  $\Delta$  in  $U'$ .

If  $w$  is not adjacent to a pendant vertex, then each of at least  $\Delta - 1$  components of  $U' - w$  contains at least two vertices, so  $n \geq 2(\Delta - 1) + 1 = 2\Delta - 1$ . Thus  $n = 2\Delta - 1$  which implies that  $U' - w$  contains exactly  $\Delta - 1$  components and each component has two vertices. It follows that  $U'$  is  $CS_{3,0,\Delta-2}$  which is a special case of  $CS_{3,2\Delta-n-1,n-\Delta-1}$  for  $n = 2\Delta - 1$ .

Now we consider the case when  $w$  is adjacent to a pendant vertex. Let  $w_0$  be a pendant vertex adjacent to  $w$  in  $U'$ .

**Claim 1.** The cycle of  $U'$  contains  $w$ .

Let  $C_k = v_1 v_2 \dots v_k v_1$ , where  $k \geq 3$ , be the unique cycle in  $U'$ . Assume to the contrary that  $w \notin V(C_k)$ . So, we can assume that  $d_{U'}(v_i) < \Delta$  for all  $i = 1, 2, \dots, k$ . Without loss of generality, suppose that  $v_1$  is the vertex of  $C_k$  closest to  $w$ . We denote the vertices adjacent to  $v_2$  in  $U'$  which are not in  $V(C_k)$  by  $u_1, u_2, \dots, u_s$ , where  $s \geq 0$ . We define  $U_1$  with  $V(U_1) = V(U')$  and  $E(U_1) = \{w_0 v_3, w_0 u_1, \dots, w_0 u_s\} \cup E(U') \setminus \{v_2 v_3, v_2 u_1, \dots, v_2 u_s\}$ . Then  $U_1$  is also a unicyclic graph. We obtain

$$\chi_a(U') - \chi_a(U_1) = (\Delta + 1)^a - [\Delta + d_{U'}(v_2)]^a + [d_{U'}(v_1) + d_{U'}(v_2)]^a - (d_{U'}(v_1) + 1)^a < 0,$$

since by Lemma 1.1, we have

$$[d_{U'}(v_1) + d_{U'}(v_2)]^a - (d_{U'}(v_1) + 1)^a < [\Delta + d_{U'}(v_2)]^a - (\Delta + 1)^a$$

for  $a < 0$ . So  $\chi_a(U') < \chi_a(U_1)$ , a contradiction.

**Claim 2.**  $w$  is the only vertex of  $U'$  which has degree greater than 2.

We prove Claim 2 by contradiction. We denote by  $u$  a vertex farthest from  $w$  such that  $d_{U'}(u) = s$ , where  $3 \leq s \leq \Delta$ . So  $u \neq w$ . Then  $U'$  contains a pendant path  $uu_1 \dots u_p$ , where  $p \geq 1$ . We define  $U_2$  with  $V(U_2) = V(U')$  and  $E(U_2) = \{w_0 u_1\} \cup E(U') \setminus \{uu_1\}$ . Then

$$\begin{aligned} \chi_a(U') - \chi_a(U_2) &= \sum_{v \in N_{U'}(u) \setminus \{u_1\}} ([d_{U'}(v) + d_{U'}(u)]^a - [d_{U_2}(v) + d_{U_2}(u)]^a) \\ &\quad + [d_{U'}(u) + d_{U'}(u_1)]^a - [d_{U_2}(w_0) + d_{U_2}(u_1)]^a + [d_{U'}(w) + d_{U'}(w_0)]^a - [d_{U_2}(w) + d_{U_2}(w_0)]^a \\ &= \sum_{v \in N_{U'}(u) \setminus \{u_1\}} [(d_{U'}(v) + s)^a - (d_{U'}(v) + s - 1)^a] \\ &\quad + [s + d_{U'}(u_1)]^a - [2 + d_{U'}(u_1)]^a + (\Delta + 1)^a - (\Delta + 2)^a \\ &< [s + d_{U'}(u_1)]^a - [2 + d_{U'}(u_1)]^a + (\Delta + 1)^a - (\Delta + 2)^a, \end{aligned}$$

since  $(d_{U'}(v) + s)^a < (d_{U'}(v) + s - 1)^a$ .

If  $p \geq 2$ , then  $d_{U'}(u_1) = 2$ . We obtain

$$\begin{aligned} \chi_a(U') - \chi_a(U_2) &< (s + 2)^a - 4^a + (\Delta + 1)^a - (\Delta + 2)^a \\ &\leq 5^a - 4^a + (\Delta + 1)^a - (\Delta + 2)^a \\ &\leq 0, \end{aligned}$$

since  $(s + 2)^a \leq 5^a$ ,  $5^a - 4^a + (\Delta + 1)^a - (\Delta + 2)^a = 0$  if  $\Delta = 3$ , and by Lemma 1.1, we get  $5^a - 4^a < (\Delta + 2)^a - (\Delta + 1)^a$  for  $a < 0$  and  $\Delta \geq 4$ .

If  $p = 1$ , then  $d_{U'}(u_1) = 1$ . We get

$$\begin{aligned} \chi_a(U') - \chi_a(U_2) &< (s + 1)^a - 3^a + (\Delta + 1)^a - (\Delta + 2)^a \\ &< (s + 2)^a - 4^a + (\Delta + 1)^a - (\Delta + 2)^a \\ &\leq 0, \end{aligned}$$

since by Lemma 1.1, we have  $(s + 1)^a - 3^a < (s + 2)^a - 4^a$ , and in the previous paragraph, it is shown that  $(s + 2)^a - 4^a + (\Delta + 1)^a - (\Delta + 2)^a \leq 0$ . Thus  $\chi_a(U') < \chi_a(U_2)$  which is a contradiction. The proof of Claim 2 is complete.

So,  $U'$  consists of a cycle containing  $w$ , and  $\Delta - 2$  pendant paths with the end vertex  $w$ . That cycle must have length 3 and every pendant path must have length at most 2. (Otherwise, if the cycle has length at least 4 or if there is a pendant path of length at least 3, then we can replace the cycle/the path by a cycle/a pendant path with one less edge, and join a new vertex to  $w_0$  to obtain a new graph  $U_3$ . Note that  $U_3$  is a unicyclic graph of order  $n$ . Then

$$\chi_a(U') - \chi_a(U_3) = 4^a - 3^a + (\Delta + 1)^a - (\Delta + 2)^a < 0,$$

since by Lemma 1.1, we have  $4^a - 3^a < (\Delta + 2)^a - (\Delta + 1)^a$  for  $a < 0$ . Thus  $\chi_a(U') < \chi_a(U_3)$  which is a contradiction.)

So  $U'$  contains  $C_3$  and  $w$  is the end vertex of  $\Delta - 2$  pendant paths. It follows that there are  $n - \Delta - 1$  vertices which are not adjacent to  $w$ , so  $w$  is the end vertex of  $n - \Delta - 1$  pendant paths having length 2 and  $2\Delta - n - 1$  pendant paths having length 1. Hence  $U'$  is  $\text{CSl}_{3,2\Delta-n-1,n-\Delta-1}$ .  $\square$

#### 4. Largest value of $\chi_a$ for $a > 0$ and smallest value of $\chi_a$ for $a < 0$

Let us present a lemma on degrees of vertices in a unicyclic graph if the number of vertices is small.

**Lemma 4.1.** Let  $U$  be a unicyclic graph with  $n$  vertices. If  $U$  contains a vertex of degree  $D$ , where  $n \leq 2D - 1$ , then no other vertex of  $U$  has degree greater than  $D$ .

**Proof.** We denote by  $w$  a vertex having degree  $D$  in  $U$ . We prove that no other vertex has degree greater than  $D$ . Assume to the contrary that  $U$  contains a vertex, say  $v$ , having degree  $D + c$ , where  $c \geq 1$ . Let  $P$  be a shortest path between  $v$  and  $w$  in  $U$ . (Note that if  $uw \in U$ , then the length of  $P$  is 1.) So,  $w$  has  $D - 1$  neighbours which are not in  $P$ , and  $v$  has at least  $D$  neighbours not in  $P$  (since  $d_U(v) \geq D + 1$ ). Since  $U$  is unicyclic,  $v$  and  $w$  have at most one common neighbour which is not in  $P$ . Therefore  $U$  contains at least  $2D - 2$  vertices not in  $P$ . Since  $P$  contains at least two vertices  $v$  and  $w$ , we have  $n \geq 2D$  which is a contradiction.  $\square$

In Lemma 4.2, we compare  $\chi_a$  of two graphs. Note that if  $G_1$  is a tree/unicyclic graph of order  $n$ , then  $G_2$  is also a tree/unicyclic graph of order  $n$ .

**Lemma 4.2.** Let  $G_1$  be a graph containing an edge  $uv$  which is not pendant and not contained in a cycle of length 3. Let us construct a graph  $G_2$  from  $G_1$  by removing the edge  $uv$ , identifying two vertices  $u$  and  $v$ , adding a vertex  $u'$  and a pendant edge  $uu'$ .

(i) If  $a > 0$ , then  $\chi_a(G_1) < \chi_a(G_2)$ .

(ii) If  $a < 0$ , then  $\chi_a(G_1) > \chi_a(G_2)$ .

**Proof.** Let  $\{v, u_1, u_2, \dots, u_{k-1}\}$  and  $\{u, v_1, v_2, \dots, v_{l-1}\}$  be the sets of vertices adjacent to  $u$  and  $v$  in  $G_1$ , respectively. So  $d_{G_1}(u) = k$  and  $d_{G_1}(v) = l$ , where  $k, l \geq 2$ . We have

$$\begin{aligned} \chi_a(G_1) - \chi_a(G_2) &= [d_{G_1}(u) + d_{G_1}(v)]^a - [d_{G_2}(u) + d_{G_2}(u')]^a + \sum_{i=1}^{k-1} [d_{G_1}(u) + d_{G_1}(u_i)]^a - \sum_{i=1}^{k-1} [d_{G_2}(u) + d_{G_2}(u_i)]^a \\ &\quad + \sum_{i=1}^{l-1} [d_{G_1}(v) + d_{G_1}(v_i)]^a - \sum_{i=1}^{l-1} [d_{G_2}(u) + d_{G_2}(v_i)]^a \\ &= (k + l)^a - (k + l - 1 + 1)^a + \sum_{i=1}^{k-1} ([k + d_{G_1}(u_i)]^a - [k + l - 1 + d_{G_1}(u_i)]^a) \\ &\quad + \sum_{i=1}^{l-1} ([l + d_{G_1}(v_i)]^a - [k + l - 1 + d_{G_1}(v_i)]^a). \end{aligned}$$

(i) For  $a > 0$ , we have  $[k + d_{G_1}(u_i)]^a < [k + l - 1 + d_{G_1}(u_i)]^a$  and  $[l + d_{G_1}(v_i)]^a < [k + l - 1 + d_{G_1}(v_i)]^a$ , thus  $\chi_a(G_1) < \chi_a(G_2)$ .

(ii) For  $a < 0$ , we have  $[k + d_{G_1}(u_i)]^a > [k + l - 1 + d_{G_1}(u_i)]^a$  and  $[l + d_{G_1}(v_i)]^a > [k + l - 1 + d_{G_1}(v_i)]^a$ , thus  $\chi_a(G_1) > \chi_a(G_2)$ .  $\square$

In Theorem 4.1, we present the unicyclic graph with the smallest  $\chi_a$  for  $a < 0$  and largest  $\chi_a$  for  $a > 0$ .

**Theorem 4.1.** Among unicyclic graphs of order  $n$  and maximum degree  $\Delta$ , where  $\lceil \frac{n+1}{2} \rceil \leq \Delta \leq n - 2$ , the graph  $CSS_{3, \Delta-2, n-\Delta-1}$  has the smallest value of  $\chi_a$  for  $a < 0$  and the largest value of  $\chi_a$  for  $a > 0$ .

**Proof.** Note that  $\lceil \frac{n+1}{2} \rceil \leq \Delta \leq n - 2$  implies that  $n \geq 5$ , and consequently  $\Delta \geq 3$ . Also  $n \leq 2\Delta - 1$ . Among unicyclic graphs of order  $n$  and maximum degree  $\Delta$ , where  $\lceil \frac{n+1}{2} \rceil \leq \Delta \leq n - 2$ , let  $U'$  be a unicyclic graph having the smallest  $\chi_a$  for  $a < 0$  (the largest  $\chi_a$  for  $a > 0$ ).

Let  $C_k = v_1 v_2 \dots v_k v_1$  be the unique cycle in  $U'$ . Let  $w$  be a vertex of degree  $\Delta$  in  $U'$ . Let  $P$  be a path between  $w$  and a vertex on  $C_k$  which is closest to  $w$ . Without loss of generality, assume that  $v_1$  is closest to  $w$  (possibly  $w = v_1$ ). Let  $S$  be the set containing all the vertices of  $C_k$  and  $P$ , so  $S = V(C_k) \cup V(P)$ .

**Claim 1.** Each vertex which is not in  $S \cup N_{U'}(w)$  has degree 1.

Assume to the contrary that there exists a vertex which is not in  $S \cup N_{U'}(w)$  with degree at least 2. We denote by  $v$  a vertex farthest from  $S \cup N_{U'}(w)$  such that  $d_{U'}(v) \geq 2$ . So  $v$  is adjacent to exactly one non-pendant vertex, say  $u$ , in  $U'$ . Note that  $u \neq w$ . Then, by Lemma 4.2, we can construct a unicyclic graph whose  $\chi_a$  is smaller for  $a < 0$  (larger for  $a > 0$ ). By Lemma 4.1, that new graph still has maximum degree  $\Delta$ . Hence, we have a contradiction.

**Claim 2.**  $k = 3$  and the length of  $P$  is at most 1.

Assume to the contrary that  $k \geq 4$  or the length of  $P$  is at least 2. Then  $U'$  contains an edge which is not pendant, not containing  $w$  and not contained in a cycle of length 3, thus by Lemma 4.2, we can construct a unicyclic graph whose  $\chi_a$  is smaller for  $a < 0$  (larger for  $a > 0$ ) which is a contradiction.

**Claim 3.** Every neighbour of  $w$  which is not in  $S$ , is a pendant vertex.

Assume to the contrary, that  $u$  is a neighbour of  $w$  which is not in  $S$  such that  $d_{U'}(u) \geq 2$ . Let  $\{w, u_1, u_2, \dots, u_{r-1}\}$  be the set of vertices adjacent to  $u$  in  $U'$ . We have  $r \geq 2$  and  $d_{U'}(u_i) = 1$  for  $i = 1, 2, \dots, r - 1$  (by Claim 1). Note that  $wv_1 \in E(U')$  if  $w \notin V(C_3)$ , and  $w = v_1$  if  $w \in V(C_3)$ . Let  $d_{U'}(v_1) = p$  and  $d_{U'}(v_2) = l$ . We have  $3 \leq p \leq \Delta$  and  $l \geq 2$ . Note that

$$l + d_{U'}(v_3) \leq \Delta + 1.$$

Otherwise, if  $d_{U'}(v_2) + d_{U'}(v_3) \geq \Delta + 2$ , then  $|N_{U'}(v_2) \cup N_{U'}(v_3)| \geq \Delta + 1$  (since  $|N_{U'}(v_2) \cap N_{U'}(v_3)| = 1$ ). Moreover,  $U'$  contains a vertex  $u_1$ , and  $w$  is adjacent to at least  $\Delta - 2$  vertices which are not in  $N_{U'}(v_2) \cup N_{U'}(v_3)$ , thus  $n \geq (\Delta + 1) + 1 + (\Delta - 2) = 2\Delta$  which is a contradiction.

We define  $U_1$  with  $V(U_1) = V(U')$  and  $E(U_1) = \{v_2u_1, v_2u_2, \dots, v_2u_{r-1}\} \cup E(U') \setminus \{uu_1, uu_2, \dots, uu_{r-1}\}$ . By Lemma 4.1, the maximum degree of  $U_1$  is  $\Delta$ . We obtain

$$\begin{aligned} \chi_a(U') - \chi_a(U_1) &= [d_{U'}(w) + d_{U'}(u)]^a - [d_{U_1}(w) + d_{U_1}(u)]^a + \sum_{i=1}^{r-1} [d_{U'}(u) + d_{U'}(u_i)]^a - \sum_{i=1}^{r-1} [d_{U_1}(v_2) + d_{U_1}(u_i)]^a \\ &\quad + \sum_{v \in N_{U'}(v_2) \setminus \{v_1, v_3\}} ([d_{U'}(v_2) + d_{U'}(v)]^a - [d_{U_1}(v_2) + d_{U_1}(v)]^a) \\ &\quad + [d_{U'}(v_1) + d_{U'}(v_2)]^a - [d_{U_1}(v_1) + d_{U_1}(v_2)]^a + [d_{U'}(v_2) + d_{U'}(v_3)]^a - [d_{U_1}(v_2) + d_{U_1}(v_3)]^a \\ &= (\Delta + r)^a - (\Delta + 1)^a + (r - 1)(r + 1)^a - (r - 1)(l + r - 1 + 1)^a \\ &\quad + \sum_{v \in N_{U'}(v_2) \setminus \{v_1, v_3\}} ([l + d_{U'}(v)]^a - [l + r - 1 + d_{U'}(v)]^a) \\ &\quad + (p + l)^a - (p + l + r - 1)^a + [l + d_{U'}(v_3)]^a - [l + r - 1 + d_{U'}(v_3)]^a. \end{aligned}$$

We consider the cases  $0 < a < 1$ ,  $a < 0$  and  $a \geq 1$  separately.

Let  $0 < a < 1$ . We have

$$\chi_a(U') - \chi_a(U_1) < (\Delta + r)^a - (\Delta + 1)^a + [l + d_{U'}(v_3)]^a - [l + r - 1 + d_{U'}(v_3)]^a \leq 0,$$

since  $(\Delta + r)^a - (\Delta + 1)^a + [l + d_{U'}(v_3)]^a - [l + r - 1 + d_{U'}(v_3)]^a = 0$  if  $l + d_{U'}(v_3) = \Delta + 1$ , and by Lemma 1.1, for  $0 < a < 1$ ,

$$(\Delta + r)^a - (\Delta + 1)^a < [l + r - 1 + d_{U'}(v_3)]^a - [l + d_{U'}(v_3)]^a$$

if  $l + d_{U'}(v_3) < \Delta + 1$ . Hence  $\chi_a(U') < \chi_a(U_1)$  which is a contradiction.

Let  $a < 0$ . We have

$$\chi_a(U') - \chi_a(U_1) > (\Delta + r)^a - (\Delta + 1)^a + [l + d_{U'}(v_3)]^a - [l + r - 1 + d_{U'}(v_3)]^a \geq 0,$$

since  $(\Delta + r)^a - (\Delta + 1)^a + [l + d_{U'}(v_3)]^a - [l + r - 1 + d_{U'}(v_3)]^a = 0$  if  $l + d_{U'}(v_3) = \Delta + 1$ , and by Lemma 1.1, for  $a < 0$ ,

$$(\Delta + r)^a - (\Delta + 1)^a > [l + r - 1 + d_{U'}(v_3)]^a - [l + d_{U'}(v_3)]^a$$

if  $l + d_{U'}(v_3) < \Delta + 1$ . Hence  $\chi_a(U') > \chi_a(U_1)$  which is a contradiction.

Let  $a \geq 1$ . If  $w = v_1$ , then  $d_{U'}(v_1) = p = \Delta$  and

$$\chi_a(U') - \chi_a(U_1) < (\Delta + r)^a - (\Delta + 1)^a + (\Delta + l)^a - (\Delta + l + r - 1)^a \leq 0,$$

since for  $a = 1$ , we have  $(\Delta + r)^a - (\Delta + 1)^a + (\Delta + l)^a - (\Delta + l + r - 1)^a = 0$ , and for  $a > 1$ , by Lemma 1.1, we have  $(\Delta + r)^a - (\Delta + 1)^a < (\Delta + l + r - 1)^a - (\Delta + l)^a$ . So  $\chi_a(U') < \chi_a(U_1)$ , a contradiction.

If  $w \neq v_1$ , then  $wv_1 \in E(U')$ . We define  $U_2$  with  $V(U_2) = V(U')$  and  $E(U_2) = \{v_1u_1, v_1u_2, \dots, v_1u_{r-1}\} \cup E(U') \setminus \{uu_1, uu_2, \dots, uu_{r-1}\}$ . By Lemma 4.1, the maximum degree of  $U_2$  is  $\Delta$ . We obtain

$$\begin{aligned} \chi_a(U') - \chi_a(U_2) &= [d_{U'}(w) + d_{U'}(u)]^a - [d_{U_2}(w) + d_{U_2}(u)]^a + \sum_{i=1}^{r-1} [d_{U'}(u) + d_{U'}(u_i)]^a - \sum_{i=1}^{r-1} [d_{U_2}(v_1) + d_{U_2}(u_i)]^a \\ &\quad + \sum_{v \in N_{U'}(v_1) \setminus \{w\}} ([d_{U'}(v_1) + d_{U'}(v)]^a - [d_{U_2}(v_1) + d_{U_2}(v)]^a) \\ &\quad + [d_{U'}(v_1) + d_{U'}(w)]^a - [d_{U_2}(v_1) + d_{U_2}(w)]^a \\ &= (\Delta + r)^a - (\Delta + 1)^a + (r - 1)(r + 1)^a - (r - 1)(p + r - 1 + 1)^a \\ &\quad + \sum_{v \in N_{U'}(v_1) \setminus \{v_2\}} ([p + d_{U'}(v)]^a - [p + r - 1 + d_{U'}(v)]^a) + (p + \Delta)^a - (p + r - 1 + \Delta)^a \\ &< (\Delta + r)^a - (\Delta + 1)^a + (p + \Delta)^a - (p + r - 1 + \Delta)^a \\ &\leq 0, \end{aligned}$$

since  $(\Delta + r)^a - (\Delta + 1)^a + (p + \Delta)^a - (p + r - 1 + \Delta)^a = 0$  if  $a = 1$ , and by Lemma 1.1, we have  $(\Delta + r)^a - (\Delta + 1)^a < (\Delta + p + r - 1)^a - (\Delta + p)^a$ . Hence  $\chi_a(U') < \chi_a(U_2)$  which is a contradiction.

**Claim 4.** The length of  $P$  is 0 (which means that  $w \in C_3$ ).

Suppose to the contrary that  $P$  has length 1, hence  $v_1w \in E(U')$ . Let  $d_{U'}(v_1) = p$ ,  $d_{U'}(v_2) = r$  and  $d_{U'}(v_3) = l$ . We have  $p \geq 3$  and  $r, l \geq 2$ . Let  $u$  be any pendant vertex adjacent to  $w$ . We obtain a new graph  $U_3$  from  $U'$

- by removing  $r - 2$  and  $l - 2$  pendant vertices which are adjacent to  $v_2$  and  $v_3$  in  $U'$ , respectively, and by joining  $r - 2 + l - 2$  new pendant vertices with  $v_1$ ,
- by removing the edge  $v_2v_3$  and adding the edge  $v_1u$ .

Note that  $U_3$  is again unicyclic having maximum degree  $\Delta$  and  $n$  vertices (by Lemma 4.1). We have

$$d_{U_3}(v_1) = p + r + l - 3, \quad d_{U_3}(v_2) = 1, \quad d_{U_3}(v_3) = 1 \quad \text{and} \quad d_{U_3}(u) = 2.$$

Then

$$\begin{aligned} \chi_a(U') - \chi_a(U_3) &= [d_{U'}(w) + d_{U'}(u)]^a - [d_{U_3}(w) + d_{U_3}(u)]^a + [d_{U'}(v_2) + d_{U'}(v_3)]^a - [d_{U_3}(v_1) + d_{U_3}(u)]^a \\ &\quad + [d_{U'}(w) + d_{U'}(v_1)]^a - [d_{U_3}(w) + d_{U_3}(v_1)]^a + [d_{U'}(v_1) + d_{U'}(v_2)]^a - [d_{U_3}(v_1) + d_{U_3}(v_2)]^a \\ &\quad + [d_{U'}(v_1) + d_{U'}(v_3)]^a - [d_{U_3}(v_1) + d_{U_3}(v_3)]^a \\ &\quad + \sum_{v \in N_{U'}(v_1) \setminus \{w, v_2, v_3\}} ([d_{U'}(v_1) + d_{U'}(v)]^a - [d_{U_3}(v_1) + d_{U_3}(v)]^a) \\ &\quad + \sum_{v \in N_{U'}(v_2) \setminus \{v_1, v_3\}} ([d_{U'}(v_2) + d_{U'}(v)]^a - [d_{U_3}(v_1) + d_{U_3}(v)]^a) \\ &\quad + \sum_{v \in N_{U'}(v_3) \setminus \{v_1, v_2\}} ([d_{U'}(v_3) + d_{U'}(v)]^a - [d_{U_3}(v_1) + d_{U_3}(v)]^a) \\ &= (\Delta + 1)^a - (\Delta + 2)^a + (r + l)^a - (p + r + l - 1)^a \\ &\quad + (\Delta + p)^a - (\Delta + p + r + l - 3)^a + (p + r)^a - (p + r + l - 2)^a \\ &\quad + (p + l)^a - (p + r + l - 2)^a + (p - 3)[(p + 1)^a - (p + r + l - 2)^a] \\ &\quad + (r - 2)[(r + 1)^a - (p + r + l - 2)^a] + (l - 2)[(l + 1)^a - (p + r + l - 2)^a]. \end{aligned}$$

For  $a > 0$ , we have  $\chi_a(U') < \chi_a(U_3)$ , and for  $a < 0$ , we have  $\chi_a(U') > \chi_a(U_3)$  which are contradictions.

**Claim 5.** One vertex of  $C_3$  is not adjacent to pendant vertices.

Suppose to the contrary that  $d_{U'}(v_2) = p$  and  $d_{U'}(v_3) = l$ , where  $p, l \geq 3$ . We obtain a new graph  $U_4$  from  $U'$  by removing  $l - 2$  pendant vertices adjacent to  $v_3$  and by joining  $l - 2$  new pendant vertices with  $v_2$ . By Lemma 4.1, the maximum degree of  $U_4$  is  $\Delta$ . We have  $d_{U_4}(v_2) = p + l - 2$  and  $d_{U_4}(v_3) = 2$ . Then

$$\begin{aligned} \chi_a(U') - \chi_a(U_4) &= [d_{U'}(w) + d_{U'}(v_2)]^a - [d_{U_4}(w) + d_{U_4}(v_2)]^a + [d_{U'}(w) + d_{U'}(v_3)]^a - [d_{U_4}(w) + d_{U_4}(v_3)]^a \\ &\quad + [d_{U'}(v_2) + d_{U'}(v_3)]^a - [d_{U_4}(v_2) + d_{U_4}(v_3)]^a \\ &\quad + \sum_{v \in N_{U'}(v_2) \setminus \{w, v_3\}} ([d_{U'}(v_2) + d_{U'}(v)]^a - [d_{U_4}(v_2) + d_{U_4}(v)]^a) \\ &\quad + \sum_{v \in N_{U'}(v_3) \setminus \{w, v_2\}} ([d_{U'}(v_3) + d_{U'}(v)]^a - [d_{U_4}(v_2) + d_{U_4}(v)]^a) \\ &= (\Delta + p)^a - (\Delta + p + l - 2)^a + (\Delta + l)^a - (\Delta + 2)^a + (p + l)^a - (p + l)^a \\ &\quad + (p - 2)[(p + 1)^a - (p + l - 1)^a] + (l - 2)[(l + 1)^a - (p + l - 1)^a] \end{aligned}$$

Let  $a \geq 1$ . We have  $(\Delta + l)^a - (\Delta + 2)^a < (\Delta + p + l - 2)^a - (\Delta + p)^a$  for  $a > 1$  by Lemma 1.1. So, for  $a \geq 1$ ,

$$(\Delta + p)^a - (\Delta + p + l - 2)^a + (\Delta + l)^a - (\Delta + 2)^a \leq 0, \quad (p + 1)^a < (p + l - 1)^a \quad \text{and} \quad (l + 1)^a < (p + l - 1)^a,$$

thus  $\chi_a(U') < \chi_a(U_4)$ .

Let  $0 < a < 1$ . We have  $(\Delta + p)^a < (\Delta + p + l - 2)^a$  and  $(l + 1)^a < (p + l - 1)^a$ . Thus

$$\chi_a(U') - \chi_a(U_4) < (\Delta + l)^a - (\Delta + 2)^a + (p + 1)^a - (p + l - 1)^a + (p - 3)[(p + 1)^a - (p + l - 1)^a] < 0,$$

since  $(\Delta + l)^a - (\Delta + 2)^a < (p + l - 1)^a - (p + 1)^a$  by Lemma 1.1. Hence  $\chi_a(U') < \chi_a(U_4)$  which is a contradiction.

Let  $a < 0$ . We have  $(\Delta + p)^a > (\Delta + p + l - 2)^a$  and  $(l + 1)^a > (p + l - 1)^a$ . Thus

$$\chi_a(U') - \chi_a(U_4) > (\Delta + l)^a - (\Delta + 2)^a + (p + 1)^a - (p + l - 1)^a + (p - 3)[(p + 1)^a - (p + l - 1)^a] > 0,$$

since  $(\Delta + l)^a - (\Delta + 2)^a > (p + l - 1)^a - (p + 1)^a$  by Lemma 1.1. Hence  $\chi_a(U') > \chi_a(U_4)$ , a contradiction.

From Claims 1–5, we know that  $U'$  consists of the cycle  $C_3$  containing the vertex  $w$  which is adjacent to  $\Delta - 2$  pendant vertices. Thus one other vertex of that  $C_3$  is adjacent to  $n - 3 - (\Delta - 2) = n - \Delta - 1$  pendant vertices. Hence  $U'$  is  $CSS_{3, \Delta-2, n-\Delta-1}$ .  $\square$

Finally, we give a solution to a part of Problem 3.1 stated in [8]. Let us define the extremal tree used in Theorem 4.2. For  $\lceil \frac{n}{2} \rceil \leq \Delta \leq n - 2$ , the double star  $S_{\Delta, n-\Delta}$  is a tree of order  $n$  with  $n - 2$  pendant vertices, where one non-pendant vertex is adjacent to  $\Delta - 1$  pendant vertices and the other non-pendant vertex is adjacent to  $n - \Delta - 1$  pendant vertices; see Fig. 5.

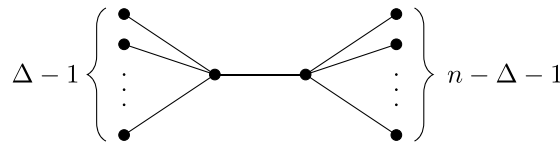
Fig. 5. Tree  $S_{\Delta, n-\Delta}$ .

Table 1

Graphs having the smallest  $\chi_a$  among unicyclic graphs of order  $n$  and maximum degree  $\Delta$  for different values of  $a$ , where  $F = (\Delta + 2)^a - (\Delta + 1)^a + 3(4^a - 5^a)$ .

| Values of $a$ | Graphs with the smallest $\chi_a$                                                                                                                                                                                                                                                                                   | Reference   |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| $a < 0$       | graph $CS_{3, \Delta-2, n-\Delta-1}$ if $\lceil \frac{n+1}{2} \rceil \leq \Delta \leq n-2$                                                                                                                                                                                                                          | Theorem 4.1 |
| $0 < a < 1$   | set $CI_{\Delta-2}^n$ if $3 \leq \Delta \leq \lfloor \frac{n+1}{2} \rfloor$<br>graph $CS_{3, 2\Delta-n-1, n-\Delta-1}$ if $\lceil \frac{n+1}{2} \rceil \leq \Delta \leq n-2$                                                                                                                                        | Theorem 2.1 |
| $a = 1$       | set $CY_{\Delta-2}^n$                                                                                                                                                                                                                                                                                               | [5]         |
| $a > 1$       | set $CPS_{\Delta-1}^n$ if $3 \leq \Delta \leq n-4$ and $F > 0$<br>set $CPS_{\Delta-1}^n \cup \{CS_{n-\Delta+2, \Delta-2}\}$ if $3 \leq \Delta \leq n-4$ and $F = 0$<br>graph $CS_{n-\Delta+2, \Delta-2}$ if $3 \leq \Delta \leq n-4$ and $F < 0$<br>graph $CS_{n-\Delta+2, \Delta-2}$ if $n-3 \leq \Delta \leq n-2$ | Theorem 2.1 |

**Theorem 4.2.** For trees with maximum degree  $\Delta$  and order  $n$ , where  $\lceil \frac{n}{2} \rceil \leq \Delta \leq n-2$  and  $\Delta \geq 3$ , the tree  $S_{\Delta, n-\Delta}$  has the largest value of  $\chi_a$  for  $0 < a < 1$ .

**Proof.** Since  $\lceil \frac{n}{2} \rceil \leq \Delta$ , we have  $n \leq 2\Delta$ . Among trees of order  $n$  and maximum degree  $\Delta$ , such that  $\lceil \frac{n}{2} \rceil \leq \Delta \leq n-2$ , we denote by  $T'$  a tree with the largest  $\chi_a$  for  $0 < a < 1$ . We denote a vertex with degree  $\Delta$  in  $T'$  by  $w$ . Note that the distance between  $w$  and any other vertex is at most 2 in  $T'$ , otherwise if there is a vertex at distance 3 from  $w$ , then  $T'$  contains an edge which is not pendant and not containing  $w$ , thus by Lemma 4.2, we can construct a tree (with maximum degree  $\Delta$  and order  $n$ ) whose  $\chi_a$  is larger for  $0 < a < 1$  which is a contradiction.

**Claim 1.** There is at most one neighbour of  $w$  which is not pendant.

Assume to the contrary that  $w_1$  and  $w_2$  are two neighbours of  $w$  such that  $d_{T'}(w_1) = p$  and  $d_{T'}(w_2) = l$ , where  $p, l \geq 2$ . Note that  $p$  and  $l$  are less than  $\Delta$ , otherwise if say  $d_{T'}(w_1) = \Delta$ , then we would get  $n \geq 2\Delta + 1$ .

We construct a new graph  $T_1$  from  $T'$  by removing  $l-1$  pendant vertices adjacent to  $w_2$  and by joining  $l-1$  new pendant vertices with  $w_1$ . Then

$$\begin{aligned}
 \chi_a(T') - \chi_a(T_1) &= (\Delta + p)^a - (\Delta + p + l - 1)^a + (\Delta + l)^a - (\Delta + 1)^a \\
 &\quad + (p-1)[(p+1)^a - (p+l)^a] + (l-1)[(l+1)^a - (p+l)^a] \\
 &< (\Delta + l)^a - (\Delta + 1)^a + (p+1)^a - (p+l)^a + (p-2)[(p+1)^a - (p+l)^a] \\
 &\leq (\Delta + l)^a - (\Delta + 1)^a + (p+1)^a - (p+l)^a \\
 &< 0,
 \end{aligned}$$

since

$$(\Delta + p)^a < (\Delta + p + l - 1)^a, \quad (l+1)^a < (p+l)^a, \quad (p+1)^a < (p+l)^a,$$

and by Lemma 1.1, we have  $(\Delta + l)^a - (\Delta + 1)^a < (p+l)^a - (p+1)^a$  for  $0 < a < 1$ . Hence  $\chi_a(T') < \chi_a(T_1)$ , a contradiction.

We know that the distance between  $w$  and any other vertex is at most 2 in  $T'$ . By Claim 1, there is at most one neighbour of  $w$  which is not pendant. It follows that  $w$  is adjacent to  $\Delta$  vertices and one of those  $\Delta$  vertices is adjacent to  $n - \Delta - 1$  pendant vertices which means that  $T'$  is  $S_{\Delta, n-\Delta}$ .  $\square$

## 5. Conclusion

In Theorems 2.1, 3.1 and 4.1, we presented graphs with the largest and smallest values of  $\chi_a$  among unicyclic graphs with order  $n$  and maximum degree  $\Delta$ . All the results given in Table 1 except for the case  $a = 1$  are our new results.

Solutions for the cases  $\Delta = 2$  and  $\Delta = n-1$  are trivial, since  $C_n$  and  $CS_{3, n-3}$  are the only unicyclic graphs for  $\Delta = 2$  and  $\Delta = n-1$  (where  $n \geq 4$ ), respectively. Therefore we do not include the cases  $\Delta = 2$  and  $\Delta = n-1$  in our tables. From Table 1, it follows that Problem 5.1 is still open.

**Problem 5.1.** Find unicyclic graphs with order  $n$  and maximum degree  $\Delta$  having the smallest value of  $\chi_a$  for  $a < 0$ , where  $3 \leq \Delta < \lceil \frac{n+1}{2} \rceil$ .

**Table 2**

Graphs having the largest  $\chi_a$  among unicyclic graphs of order  $n$  and maximum degree  $\Delta$  for different values of  $a$ .

| Values of $a$   | Graphs with the largest $\chi_a$                                                                                                                                            | Reference   |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| $a < -1$        | graph $CSl_{3,2\Delta-n-1,n-\Delta-1}$ if $\lceil \frac{n+1}{2} \rceil \leq \Delta \leq n-2$                                                                                | Theorem 3.1 |
| $-1 \leq a < 0$ | set $Cl_{\Delta-2}^n$ if $3 \leq \Delta \leq \lfloor \frac{n+1}{2} \rfloor$<br>graph $CSl_{3,2\Delta-n-1,n-\Delta-1}$ if $\lceil \frac{n+1}{2} \rceil \leq \Delta \leq n-2$ | [7]         |
| $a > 0$         | graph $CSS_{3,\Delta-2,n-\Delta-1}$ if $\lceil \frac{n+1}{2} \rceil \leq \Delta \leq n-2$                                                                                   | Theorem 4.1 |

Our new results and other known results on the largest  $\chi_a$  are given in Table 2. It remains to solve the cases stated in Problem 5.2.

**Problem 5.2.** Find unicyclic graphs with order  $n$  and maximum degree  $\Delta$  having the largest value of  $\chi_a$  for  $a < -1$  and  $a > 0$ , where  $3 \leq \Delta < \lceil \frac{n+1}{2} \rceil$ .

Note that if  $a = 0$ , then for every unicyclic graph  $U$ , we obtain

$$\chi_0(U) = \sum_{uv \in E(U)} [d_U(u) + d_U(v)]^0 = \sum_{uv \in E(U)} 1 = |E(U)| = n.$$

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### Data availability

No data was used for the research described in the article.

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